| Module | Problem | Candidates (time complexity) | Selection | Justification |
|---|---|---|---|---|
| **A. Relief inventory** | **Input:** n crates (integer weight 1-20 kg, value after the day's multiplier), capacity W. **Output:** a subset of indivisible crates with total weight <= W. **Objective:** maximise total value. | 1. 0/1 knapsack DP over capacity: O(nW) time, O(nW) bits to rebuild the set (O(W) values).<br>2. Greedy by value/weight ratio (exact fraction compare), skip misfits: O(n log n).<br>3. Exhaustive search or branch and bound: $O(2^n)$ worst case. | **Primary:** knapsack DP. **Baseline:** ratio greedy (also exhaustive search on tier S as a correctness reference). | **Data property:** weights are small integers, so W <= 0.7 x 20n = 14n. The pseudo-polynomial DP is then about $O(n^2)$ (about $3x10^7$ cells for n = 1,500) and exact. **Assumption:** W stays polynomial in n. In our data greedy is not optimal in 20/20 tier S cases and reaches at most 60% of the optimum in trap 47, so an exact method is needed. **Fails when:** weights or W become very large (table memory O(nW)); then use meet-in-the-middle or branch and bound. The daily value multipliers change values only, so the DP is unaffected. |
| **B. Route planning** | **Input:** V locations, E roads, source 0, today's blocked roads. Directed step cost u to v = minutes + penalty(v), always >= 1. **Output:** cheapest route and cost or 'no route'; set of reachable locations; fewest-roads route (ascending-id tie rule). **Objective:** minimum cost. | 1. Dijkstra with a binary heap: O((V+E) log V).<br>2. Bellman-Ford: O(VE).<br>3. BFS (level by level, neighbours in ascending id): O(V+E); gives reachability and fewest roads, but ignores cost.<br>4. Dijkstra with an array scan: $O(V^2)$. | **Primary:** heap Dijkstra for cheapest routes plus BFS for reachable set and fewest-roads route. **Baseline:** Bellman-Ford with early exit (cost), and 'fewest-roads route used as the cheapest' to measure the cost gap. | **Data property:** all step costs are positive and the maps are sparse (E <= 3V), so heap Dijkstra O(E log V) beats Bellman-Ford O(VE) by a large factor at V up to 3,000. Trap 48 has fewest-roads routes costing at least 1.5x a cheapest route, so a cost-blind BFS is wrong and cost-aware search is required. **Assumption:** costs are non-negative and the graph is sparse. **Fails when:** a negative cost appears (Dijkstra is incorrect; use Bellman-Ford), or the graph is dense (E near $V^2$), where the array version $O(V^2)$ is faster. Blocked roads that disconnect the target (study 5) are returned as 'no route', not as an error. |
| **C. Rescue scheduling** | **Input:** n windows (start < finish in minutes, value 1-100); on each day windows whose id + day is divisible by 5 are removed. **Output:** pairwise compatible windows (touching is compatible). **Objective:** most windows, or most total value. | 1. Earliest-finish-first greedy: O(n log n), exact for the count.<br>2. Weighted interval DP, sort by finish and binary search for the last compatible window: O(n log n).<br>3. Greedy by highest value: O(n log n).<br>4. Exhaustive subset search: $O(2^n)$. | **Primary:** earliest-finish greedy for the count and weighted interval DP for the value. **Baseline:** greedy by highest value (exhaustive search on tier S as reference). | **Data property:** times are integers in 0-1440 and ties are broken by window_id, so one sorted order gives a deterministic answer. Study 3 has short windows where each overlaps at least 2 others on average, so taking the highest value first blocks several good windows and is not optimal, while the DP stays exact. **Assumption:** one crew, so a window is compatible only if start >= the previous finish. **Fails when:** several crews can work in parallel or windows need set-up time between them; then the problem becomes a different (machine-scheduling) problem and both algorithms stop being exact. |
| **D. Emergency queue** | **Input:** m alarms (tick, priority 1-10, alarm_id), plus the added alarms of days 2 and 5 at tick 0. **Output:** release order, one alarm per tick, chosen among alarms that have arrived, by urgency (priority desc, tick asc, alarm_id asc). **Objective:** always release the most urgent waiting alarm. | 1. Binary heap priority queue: O(m log m) in total.<br>2. Sorted list kept in order with insertion: $O(m^2)$ element shifts.<br>3. Bucket queue with one FIFO per priority (10 levels): O(m) with constant 10. | **Primary:** binary heap with the full 3-part urgency key. **Baseline:** sorted-list insertion (and a linear scan for the maximum). | **Data property:** on tiers M and L alarms arrive at 1.2-1.8 per tick but only one is released per tick, so the waiting queue grows to hundreds or thousands of alarms. The heap costs O(log q) per operation against O(q) for the list, and study 4 (very high rate) and study 2 (bursts) are the worst cases. **Assumption:** a long backlog and a total order on keys. **Fails when:** the backlog stays tiny (arrival rate below 1 per tick); then the list is as fast as the heap and simpler. With only 10 priority levels the bucket queue could beat the heap, which we will test. |

| Member | Contribution |
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| Muhammad Umar<br>22I-0644, Section AI-A | **Data:** generator modules for the cases row and crates, the road network (locations, roads, waves) and the four trap cases (46-49).<br>**Validator:** checks V3 and V4 (identifiers, references, connectivity, waves), V8 (trap rules) and the cross-case variety rules; the test runner that builds the broken copies.<br>**Documents:** approach note part 3 (the traps).<br>**Iteration 2 modules proposed:** A (Relief inventory) and the seven-day campaign coordinator. |
| Haider Ali<br>22I-0502, Section AI-A | **Data:** generator modules for radio (log, patterns, reports), windows, incidents and the Tier S claims by exhaustive search.<br>**Validator:** checks V1 and V2 (sheets, headers, types, ranges) and V5 (pattern counts, edit_bound, claims).<br>**Documents:** approach note part 1 (from story to numbers) and the generator README.<br>**Iteration 2 modules proposed:** B (Route planning). |
| Awwab Ahmed<br>23I-0079, Section AI-A | **Data:** generator modules for sites and the planted tour, pipeline dimensions, drum prices, sensors, safety rules and rations.<br>**Validator:** checks V6 (planted objects) and V7 (signature rules); the validation report.<br>**Documents:** approach note parts 2 and 4 (planted objects, what will be hard) and this proposal.<br>**Iteration 2 modules proposed:** C (Rescue scheduling) and D (Emergency queue). |